Exp 7 : Analysis of 5G NR Frame Structures and Subcarrier Spacing

Aim :

To analyze the 5G NR frame structure by studying subcarrier spacing, slot duration, OFDM symbols and simplified downlink/uplink resource assignments using MATLAB, and to relate the radio resources to gNodeB functions and RRC operation.

Objective 1 - Generate / Inspect a Simplified 5G NR Resource Grid :

Code :

```
clc;
clear;
close all;

%% 1. Define OFDM Resource Grid Parameters
Nsc = 24;     % Number of subcarriers
Nsym = 14;    % Number of OFDM symbols per slot

% Initialize resource grid
gridData = zeros(Nsym, Nsc);

%% 2. Assign Downlink and Uplink Resources

% Downlink (DL): OFDM symbols 1 to 10
gridData(1:10, :) = 1;

% Uplink (UL): OFDM symbols 11 to 14
gridData(11:14, :) = 2;

%% 3. Plot the Time-Frequency Resource Grid
figure('Color', 'w');

imagesc(1:Nsc, 1:Nsym, gridData);

set(gca, 'YDir', 'normal');

xlabel('Subcarrier Index');
ylabel('OFDM Symbol Index');
title('Simplified 5G NR Time-Frequency Resource Grid');

xticks(1:2:Nsc);
yticks(1:Nsym);

colormap([0.85 0.92 1.00; 0.30 0.65 0.90; 1.00 0.65 0.30]);
clim([0 2]);
```

```
cb = colorbar;
cb.Ticks = [0 1 2];
cb.TickLabels = {'Unused', 'Downlink (DL)', 'Uplink (UL)'};

grid on;
set(gca, 'GridColor', 'k', 'GridAlpha', 0.15);

%% 4. Display Resource Allocation Information
fprintf('Simplified 5G NR Resource Grid\n');
```

```
Simplified 5G NR Resource Grid
```

```
fprintf('------------------------------\n');
```

```
----------------------------
```

```
fprintf('Number of subcarriers: %d\n', Nsc);
```

```
Number of subcarriers: 24
```

```
fprintf('OFDM symbols per slot: %d\n', Nsym);
```

```
OFDM symbols per slot: 14
```

```
fprintf('Downlink symbols: 1-10\n');
```

```
Downlink symbols: 1-10
```

```
fprintf('Uplink symbols: 11-14\n');
```

```
Uplink symbols: 11-14
```

```
fprintf('Total resource elements: %d\n', Nsc * Nsym);
```

```
Total resource elements: 336
```

```

%% 5. Display Execution Date and Time
currentTime = datetime('now', ...
    'Format', 'dd-MM-yyyy HH:mm:ss');

fprintf('\nExecution Date and Time: %s\n', ...
    char(currentTime));
```

```
Execution Date and Time: 20-10-2026 10:01:40
```

Objective 2 - Compare SCS and Slot Duration

Code :

```
clc;
clear;
close all;

%% 1. Define 5G NR Numerology Parameters
scs = [15 30 60 120 240];       % Subcarrier spacing (kHz)
```

```
mu = 0:4;                          % Numerology index
slot = [1 0.5 0.25 0.125 0.0625]; % Slot duration (ms)

%% 2. Plot Slot Duration vs Subcarrier Spacing
figure('Color', 'w');

plot(scs, slot, 'o-', 'LineWidth', 2, ...
    'MarkerSize', 7);

grid on;
xlabel('Subcarrier Spacing (kHz)');
ylabel('Slot Duration (ms)');
title('5G NR Slot Duration for Different Subcarrier Spacings');

xticks(scs);

%% 3. Plot Subcarrier Spacing vs Numerology Index
figure('Color', 'w');

bar(mu, scs);

grid on;
xlabel('Numerology Index (\mu)');
ylabel('Subcarrier Spacing (kHz)');
title('5G NR Subcarrier Spacing for Different Numerologies');

xticks(mu);
yticks(scs);
ylim([0 260]);

%% 4. Display Results
fprintf('5G NR Numerology Results\n');
```

```
5G NR Numerology Results
```

```
fprintf('------------------------\n');
```

```
------------------------
```

```

for k = 1:length(mu)
    fprintf('Numerology index: %d | SCS: %d kHz | Slot duration: %.4f ms\n',
...
        mu(k), scs(k), slot(k));
end
```

```
Numerology index: 0 | SCS: 15 kHz | Slot duration: 1.0000 ms
Numerology index: 1 | SCS: 30 kHz | Slot duration: 0.5000 ms
Numerology index: 2 | SCS: 60 kHz | Slot duration: 0.2500 ms
Numerology index: 3 | SCS: 120 kHz | Slot duration: 0.1250 ms
Numerology index: 4 | SCS: 240 kHz | Slot duration: 0.0625 ms
```

```
%% 5. Display Execution Date and Time
currentTime = datetime('now', ...
    'Format', 'dd-MM-yyyy HH:mm:ss');

fprintf('\nExecution Date and Time: %s\n', char(currentTime));
```

```
Execution Date and Time: 20-10-2026 10:01:40
```

Objective 3 : Identify Baic DL and UL Resource Assignments :

Code :

```
clc;
clear;
close all;

%% 1. Define OFDM Symbols and Resource Allocation
symbols = 1:14;

% Downlink allocation: Symbols 1 to 10
DL = [1 1 1 1 1 1 1 1 1 1 0 0 0 0];

% Uplink allocation: Symbols 11 to 14
UL = [0 0 0 0 0 0 0 0 0 0 1 1 1 1];

%% 2. Plot Downlink and Uplink Assignments
figure('Color', 'w');

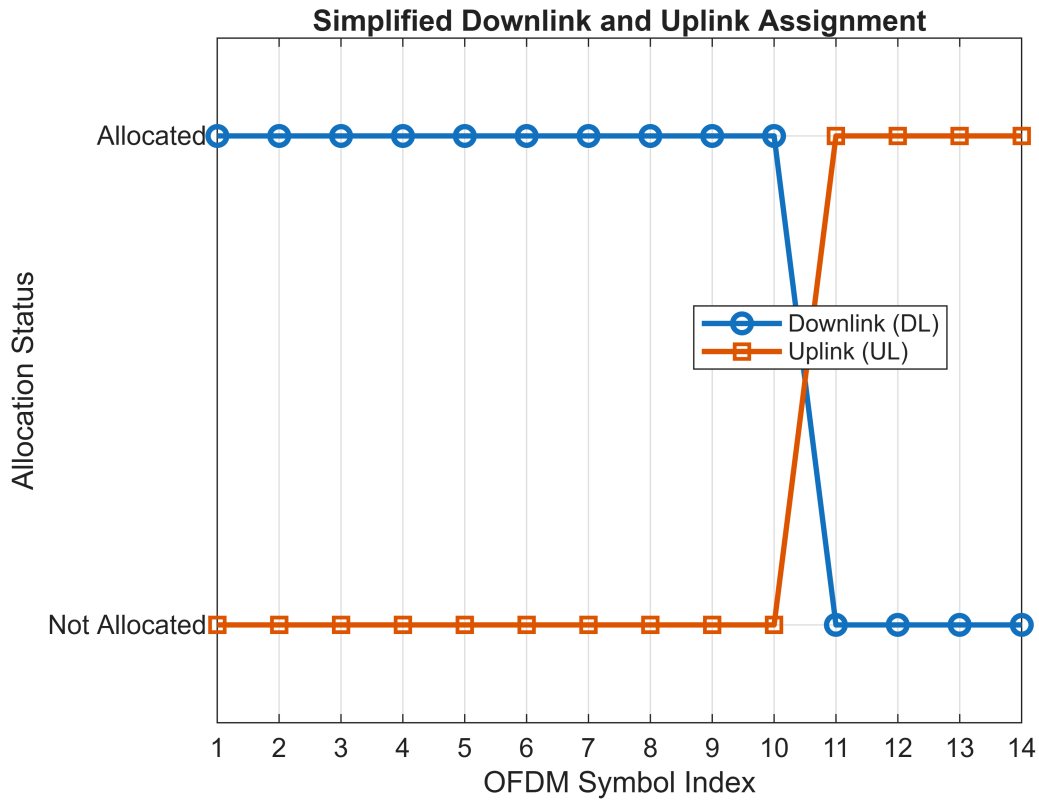
plot(symbols, DL, 'o-', 'LineWidth', 2, ...
    'MarkerSize', 7);
hold on;

plot(symbols, UL, 's-', 'LineWidth', 2, ...
    'MarkerSize', 7);

grid on;
xlabel('OFDM Symbol Index');
ylabel('Allocation Status');
title('Simplified Downlink and Uplink Assignment');

legend('Downlink (DL)', 'Uplink (UL)', ...
    'Location', 'best');

xticks(symbols);
yticks([0 1]);
yticklabels({'Not Allocated', 'Allocated'});
xlim([1 14]);
ylim([-0.2 1.2]);
```




```
%% 3. Display Resource Allocation Information
fprintf('5G NR Downlink and Uplink Resource Assignment\n');
```

```
5G NR Downlink and Uplink Resource Assignment
```

```
fprintf('-----------------------------------------\n');
```

```
-----------------------------------------
```

```
fprintf('Total OFDM symbols: %d\n', length(symbols));
```

```
Total OFDM symbols: 14
```

```
fprintf('Downlink symbols: 1-10\n');
```

```
Downlink symbols: 1-10
```

```
fprintf('Uplink symbols: 11-14\n');
```

```
Uplink symbols: 11-14
```

```
%% 4. Display Execution Date and Time
currentTime = datetime('now', ...
    'Format', 'dd-MM-yyyy HH:mm:ss');

fprintf('\nExecution Date and Time: %s\n', char(currentTime));
```

Execution Date and Time: 20-10-2026 10:01:40

Objective 4 : Relate Resources to gNodeB, RAN and RRC :

The gNodeB is the principal 5G NR radio access node. It participates in radio-resource management and scheduling and provides the radio interface between the UE and the 5G core. RRC is a control-plane protocol responsible for radio-resource configuration and connection-related control. The simplified grid therefore represents the radio resources that are scheduled and configured within the RAN; the exact allocation in a live network depends on traffic, channel conditions, QoS and scheduling decision

Result :

1. A simplified 5G NR resource grid was generated and inspected.
2. The effect of different SCS values on slot duration was analyzed.
3. Basic downlink and uplink resource assignments were represented.
4. The relationship among NR radio resources, gNodeB operation, RAN organization and RRC was explained.